

ENGLISH HOUSING SURVEY (EHS)

Housing Stock Dataset User Guide

2019

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Introduction to the English Housing Survey (EHS)

- 1 The EHS is a national survey commissioned by the Department for Levelling Up, Housing and Communities (DLUHC) that collects information about people's housing circumstances and the condition and energy efficiency of housing in England.
- 2 The annual survey has a complex multi-stage methodology consisting of 2 main elements: an initial interview survey of around 13,300 households with a follow up physical inspection of a sub-sample of approximately 6,200 of these dwellings, including vacant dwellings.
- 3 In 2011-12, a cost review of the EHS resulted in several changes to the survey. The main changes to the survey were:
 - a reduction to the EHS sample size
 - a reduction in the questionnaire content. This included the removal of the EHS from the Integrated Household Survey (IHS)
 - the cessation of the desk-based market valuation of sub-sampled properties

Information on these changes can be found in the document [“Proposals arising from a cost review of the English Housing Survey”](#).

- 4 The English Housing Survey comprises of:
 - An initial interview with approximately 13,300 households a year. This is referred to as the ‘full interview sample’. This dataset is made available on a single financial year basis as the EHS household dataset.
 - A follow up physical inspection of around 6,200 of these respondents’ homes to assess the condition and energy performance of the property. This is referred to as the ‘dwelling sample’. This dataset is made available on a rolling two-year basis (around 12,400 dwellings). The ‘dwelling sample’ comprises all cases where an occupied dwelling has an interview and physical survey completed plus vacant dwellings with a physical survey only.
- 5 There is a range of information that will support anyone wishing to make use of the survey data to conduct their own analysis or wishing to understand in detail how the survey is run and managed. All users are encouraged to familiarise themselves with the [technical background](#) before undertaking any detailed work using the EHS results.

Reporting

6 The reports published using 2019-20 EHS data are as follows:

EHS Headline Report – the EHS Headline Report is usually published in the December/January following the completion of fieldwork. This short report presents preliminary headline findings and is in two sections. The first section, on households, includes key indicators such as trends in tenure and household composition, overcrowding and under-occupation, housing costs, affordability and personal wellbeing. The second section, on homes, includes topics such as dwelling typology, condition, energy efficiency and smoke and gas detectors. Analysis is at a national level from both the full and dwelling samples. This is followed by separate annual reports, later in the year.

The following annual reports are being published in July 2021:

- a) **Home adaptations report:** This report focuses on households containing one or more person with a long-term limiting disability. It includes analysis of whether these households perceive their homes to be suitable for their needs, what adaptations these households require and what adaptations they currently have. It also investigates the types of households that do not have the required adaptations and why they do not have them.
- b) **Energy report:** This report examines the energy efficiency of the English housing stock. It includes analysis of: EPC ratings; energy costs; questions about Smart Meters; energy efficiency measures (such as heating systems, primary heating controls, insulation and ventilation); energy improvement works undertaken by owner occupiers; costs to make homes more energy efficient.
- c) **Home ownership report:** This report provides a profile of owner occupied households, looking at their demographics and economic circumstances. It also looks at levels of overcrowding and under-occupation, satisfaction, and housing costs and affordability including mortgage length and mortgage deposits. It then explores the housing history of these households including length of residence and moving expectations. The final chapter examines dwelling condition and energy efficiency of the stock.
- d) **Private rented sector report:** This report focuses on households and housing in the private rented sector. It covers the profile of private renters, levels of overcrowding and under-occupation, their housing costs and affordability, their housing history and housing aspirations. The report also examines the energy efficiency and dwelling condition of the private rented stock.
- e) **Social rented sector report:** This report describes the profile of households living in the social rented sector. It also covers housing satisfaction, housing costs and affordability, housing history and housing aspirations, including moves in, out and within this tenure

alongside reasons for moving. Energy efficiency and dwelling condition of the social rented stock complete this analysis.

- f) **Wellbeing & Neighbourhoods report:** The report provides a detailed overview of people's well-being and perceptions of their neighbourhood across all tenures. It includes analysis of well-being and loneliness, neighbourhood satisfaction and attitudes, as well as covering whether well-being varies according to the social environment of the neighbourhood.

After publication, the Annual Reports are available to download from the [EHS pages on the Gov.uk website](#). The reports include an overview of the survey methodology and a glossary of terms.

- 7 Alongside the annual reports and its tables and figures, the EHS also produces **Web Tables**. An updated set of tables are produced each year and made available on the [EHS pages on the GOV.UK website](#).

EHS datasets

- 8 The EHS datasets released to external users through the UK Data Service reflect the above EHS reporting strategy. Two separate datasets are therefore available as follows:
 - a) **EHS Household Dataset** –The dataset comprises the full sample primary ‘raw’ interview survey data (available via the Special Licence only), plus associated derived variables (available via both the End User Licence and the Special Licence) for all cases where an interview has been completed – around 13,300 households per annum (approximately 17,000 per annum before the 2011-12 EHS cost review). Datasets are provided for single financial years together with annual weights. This dataset should be used for any analysis where only information from the household interview is required.
 - b) **EHS Housing Stock Dataset** – The Housing Stock Dataset should be used for any analysis requiring information relating to the physical characteristics and energy efficiency of the housing stock. The dataset comprises the paired sample primary ‘raw’ interview survey and physical survey data (available via the Special Licence only), plus associated derived variables (available via both the End User Licence and the Special Licence) for all cases where a physical survey has been completed. Occupied cases comprise of data from the household interview as well as data from the physical survey. For vacant properties, only data for the physical survey are provided.

The housing stock data are made available for a two-year rolling sample with the appropriate two-year weights. For example, the EHS Housing Stock Dataset is for ‘2019’ covering the period April

2018 to March 2020. This dataset comprises of up to 12,400 cases (previously 16,500 prior to the EHS cost review). This means that users who use more than one housing stock dataset must use either odd or even years. For example, use the Housing Stock Dataset for '2017' and '2019' or '2016' and '2018', but not the dataset for '2018' and '2019' as that would double-count the cases surveyed between April 2018 and March 2019.

- 9 **This User Guide relates solely to the EHS Housing Stock Dataset.** Users who wish to access the EHS Household dataset should refer to the separate EHS Household User Guide that accompanies that dataset. DLUHC has deposited the data in this way to provide greater continuity for users of the predecessors of the EHS, the Survey of English Housing (SEH) and English House Condition Survey (EHCS).
- 10 DLUHC reviews the scope of the EHS every year so the EHS datasets deposited in the Data Archive are not identical year on year. Each year, new data files may be created, some variables may be dropped and a number of new variables may be introduced to some data files. This applies particularly to the primary 'raw' data files and users will need to be particularly careful when using these files. Further information is available in the documentation of the questionnaire and physical survey form published on the [EHS pages on the GOV.UK website](#).

Overview of the EHS Housing Stock Dataset

- 11 The EHS Housing Stock data are released in SPSS, STATA and TAB format through the [UK Data Service](#), as a download. To download or access the EHS data you will need to register with UK Data Service, though it is possible to download descriptions of data and a general overview of the survey without registering.
- 12 Two versions of the EHS Housing Stock data are released via UK Data Service; an End User Licence version and a Special Licence version which provides more detailed information. The datasets released under the different licences vary by the level of detail of the data and the measures of statistical disclosure control that have been applied. Which licence is required depends on the nature of the project using the EHS datasets. The level of licence needed to access different datasets is provided in the next section.

EHS database

- 13 Once you have downloaded the EHS data in the preferred format, you will see a number of files within WinZip. The data files in SPSS, Stata or TAB format are listed under 'spss', 'stata' or 'tab' respectively.

- 14 The EHS housing stock Special Licence dataset consists of primary 'raw' data files containing basic survey data collected by the surveyors and interviewers. These include files such as amenity_SL.sav, windows_SL.sav and numflats_SL.sav from the physical survey and attitudes_SL.sav, disability_SL.sav and employment_SL.sav from the interview survey.

Further information on content of the raw interview variable files and the raw physical variables files is provided in Appendix A and Appendix B respectively.

- 15 The EHS housing stock Special Licence datasets also include a number of files containing 'derived' variables. These are variables created by modelling or recoding the basic survey data. The Special Licence derived files include physical 18+19_SL.sav, interview 18+19_SL.sav and general 18+19_SL.sav. The Special Licence derived data files contain additional variables to that of the End User Licence derived data files. There are also further detailed files such as actual costs 18+19_SL.sav, energy performance 18+19_SL.sav and HHSRS 18+19_SL.sav.

Further information on content of these derived variable files is provided in Appendix C.

- 16 The EHS housing stock End User Licence dataset consists of only the three files containing 'derived' variables on the database; physical 18+19_EUL.sav, interview 18+19_EUL.sav and general 18+19_EUL.sav. These are variables created by modelling or recoding the basic survey data.

Further information on content of these derived variable files is provided in Appendix C.

- 17 Users are encouraged wherever possible to make use of derived files for their analysis. This is because although the primary 'raw' physical and interview survey files have been checked for dimensional consistency and plausibility, they have not been through the rigorous editing, imputation and modelling that the derived variables have. It is advised that researchers who use the primary 'raw' data files to conduct analysis should treat those files with more caution.

- 18 The second group of files, listed under 'mrdoc' are the **documentation and reports**. A number of documents are included on the EHS housing stock database to help users understand how the database is organised and which data files to use.

- 19 This includes the **interview survey documentation** which shows all the questions asked of households, the variable names and available response categories and showcards. Some new questions have been introduced and others have been dropped compared with previous years so the 2019-20 interview questionnaire provided indicates the

new interview content. Certain topics are only collected on a rotating basis.

- 20 Also available on the dataset are the **physical survey documentation**. This includes a modified pdf version of the survey form to show the variable names alongside each question; surveyor instructions and the guidance that normally appear on the facing pages of the paper survey form.
- 21 For 2019-20, the latest version of the **data dictionary** has been produced to provide comprehensive details for all the Special Licence and End User Licence variables on the corresponding generalfs19.sav, general 18+19.sav, interviewfs19.sav, interview 18+19.sav and physical 18+19.sav files. For information on the variables shown in the primary 'raw' data files, please refer to interview survey and physical survey documentation.
- 22 Included in this group of files on the zip drive is a **tool** to check whether your comparisons are statistically significant.
- 23 Additional documents have also been produced to provide comprehensive information on the detailed derived files: actual costs 18+19_SL.sav, dimensions 18+19_SL.sav, energy performance 18+19_SL.sav, 18+19 paired sample equivalised income_SL.sav, 2019 full sample equivalised income_SL.sav, HHSRS 18+19_SL.sav and standardised costs 18+19_SL.sav.
- 24 **Fuel poverty documentation** – one of the EHS datasets on the UK Data Service website covers fuel poverty. The link below provides further information relating to Energy Efficiency and Fuel Poverty.
<https://www.gov.uk/government/organisations/department-of-energy-climate-change/series/fuel-poverty-statistics>

Other sources of information

- 25 Further information about the way in which the survey is organised is available on the [EHS pages on the Gov.uk website](#).
- 26 As mentioned earlier in this note, all users are encouraged to familiarise themselves with the technical background found in the [EHS Technical Report](#) before undertaking any detailed work using the EHS data. These provide details of how the fieldwork is organised, how the sample is drawn and the results grossed each year, estimates of sampling and measurement error together with a full glossary of terms and details of how derived measures such as energy efficiency and income are created.

Using the database - first steps

27 To use the dataset correctly you must first ensure that you familiarise yourself with and be aware of the following:

- How the dataset has been produced
- What information is published
- The key concepts and definitions used in the survey
- How to locate the information you want in the database (find both survey and computed data by variable and file names)
- How to match files
- How to apply grossing factors to obtain national estimates
- How to deal with missing data
- How to deal with sampling and survey error

Each of these is described briefly in the sections below but please note that this guide is not intended to be a comprehensive source of information. Reference is made to the more detailed explanations and guidance contained in other files, publications and documents which are either on the UKDS or on the [EHS pages on the Gov.uk website](#).

The guide assumes knowledge and experience of using SPSS.

How the housing stock dataset has been produced

28 The EHS is a set of interrelated data from household and physical surveys based on a *sample of dwellings*.

29 The following describes the components of the whole EHS:

- A questionnaire-based survey of households living in the sampled dwellings which provides information on household characteristics, resources, housing costs and views of respondents about their home and neighbourhood. This is known as the **interview survey and is based on households**.
- A physical assessment of a sub-set of the sampled dwellings where permission/access was available or which were vacant at the time the interviewer called. This provides information on dwelling characteristics and a professional assessment of dwelling and neighbourhood conditions. This is known as the **physical survey and is based on dwellings**.

30 More detail on how the two surveys are conducted can be found in the [EHS Technical Report](#).

Sample structure in 2019-20

Annual sample size	28,238
Frequency of fieldwork	2 months per quarter
Number of months of interview fieldwork	12
Coverage	England only
Clustering	Partial
Design description	Two-stage sample of addresses
Achieved response rate in 2019-20	51%
- Household interviews achieved	13,332
- Paired cases achieved	6,014 (including additional 150 productive vacant cases)

31 In order to achieve the number of interviews and surveys described above, an initial sample of 45,000 addresses was drawn from the Postcode Address File (PAF). These addresses were drawn as a systematic two-stage random sample from the Royal Mail's Small User PAF.

32 For the first stage of sampling, the whole of England was grouped into 1,808 geographic 'merged areas' (clusters). These areas were generated by combining neighbouring Lower Layer Super Output Areas (LSOAs) within region, so that each cluster contained about 12,500 addresses. The 1,808 clusters were then stratified and one half (904) was randomly allocated to "Year 1" and the other half (904) to "Year 2". The allocation of clusters to years was fixed since 2012-13: "Year 1" clusters were used for 2012-13, 2014-15, 2016-17 and 2018-19, and "Year 2" clusters for 2013-14, 2015-16, 2017-18 and 2019-20.

The second stage of sampling involved selecting a stratified systematic random sample of 45,000 addresses within the 904 clusters selected for the 2019-20 sample.

33 The advantage of this two-stage approach is that it reduces the fieldwork area to half the country so interviewer and surveyor travel time / costs are significantly reduced. The disadvantage is that for any single survey year the survey is partially clustered, which will result in a small loss in statistical efficiency. However, when analysing any two years of survey data the combined sample is entirely unclustered.

For each sampled address, the predominant tenure within the postcode that contained that address was identified using Experian's Residata classifications and attached to the record.

To ensure that there is a sufficiently large sample of social renters for analysis, sub-sampling was carried out by grouping the addresses into the four housing tenure types and sub-sampling at the rates of 48% for owner occupied, 93% for private rented and 94% for social rented. As a result, 28,975 addresses were issued to interviewers.

- 34 Full details of sample methodology can be found in the [EHS Technical Report](#) on the EHS pages on Gov.uk.

Datasets released from the UK Data Service (UKDS)

- 35 The EHS is a National Statistics series. This means the statistics meet user needs, are produced, managed and disseminated to high standards, and managed impartially and objectively in the public interest.
- 36 All data files in the EHS housing stock dataset contain two years' worth of data. This means that if you use more than one housing stock dataset, you must use either odd or even years. For example, you need to use the Housing Stock Dataset for '2016' and '2018' or '2017' and '2019', but not the dataset for '2018' and '2019' as you would double-count the cases surveyed between April 2018 and March 2019. The following two-year datasets are released by the UKDS.

EHS housing stock dataset	Annual data contents	Mid point
EHS2008	2007-08 and 2008-09	April 2008
EHS2009	2008-09 and 2009-10	April 2009
EHS2010	2009-10 and 2010-11	April 2010
EHS2011	2010-11 and 2011-12	April 2011
EHS2012	2011-12 and 2012-13	April 2012
EHS2013	2012-13 and 2013-14	April 2013
EHS2014	2013-14 and 2014-15	April 2014
EHS2015	2014-15 and 2015-16	April 2015
EHS2016	2015-16 and 2016-17	April 2016
EHS2017	2016-17 and 2017-18	April 2017
EHS2018	2017-18 and 2018-19	April 2018
EHS2019	2018-19 and 2019-20	April 2019

The number of (unweighted) cases in the housing stock dataset each year is as follows:

EHS housing stock dataset	Unweighted cases, dwellings	Unweighted cases, households
EHS2008	16,150	15,523
EHS2009	16,150	15,512
EHS2010	16,670	16,047
EHS2011	14,951	14,386
EHS2012	12,763	12,269
EHS2013	12,498	12,008
EHS2014	12,297	11,851
EHS2015	12,351	11,955
EHS2016	12,292	11,924
EHS2017	12,320	11,963
EHS2018	12,562	12,203

EHS2019	12,300	11,974
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Levels of data

- 37 Most of the data is at household or dwelling level. However, some of the primary 'raw' data files contain data at sub-household and sub-dwelling level. These are indicated for the interview survey at Appendix A and for the Physical Survey at Appendix B. All the files containing derived variables are at dwelling or household level.

Types of data

- 38 **Primary 'raw' survey data (only available under Special Licence)** – primary data collected in the field are often very specific and detailed and in many cases are only collected in order to provide the building blocks for computing more useful pieces of information – in particular the household composition and income measures used in the analysis of the survey. There will also be a small amount of missing data in the primary data collected in the field.

In many cases omissions or inconsistencies in the base variables will have been identified in the course of producing derived variables for final analysis. **Users are recommended wherever possible to make use of the final derived variables (available under both the End User Licence and the Special Licence) rather than the raw data from the primary files as these are likely to contain more complete information on key topics such as rent, mortgage payments and income.**

- 39 **Derived/computed data (available under both the End User Licence and the Special Licence)** – while the information that is collected in the field provides all the base variables needed, a considerable amount of further processing and modelling is needed to fully validate the results and produce secondary variables needed for detailed analysis. Producing derived variables often involves reconciliation of conflicting information from different parts of the survey (e.g. tenure) and/or the imputation of missing data where this has been possible from other data collected in the survey.
- 40 It is important to note that changes introduced to some of the EHS modules will have a significant impact on any time series analysis and users should familiarise themselves with the relevant [EHS Technical background](#) before drawing conclusions about changes over time. Revised variable names have been used wherever there has been a major break in methodology year on year to minimise the risk of data being misinterpreted.

- 41 **Missing data** – as with all surveys there will be some level of missing data on the primary ‘raw’ data files where information was not available or for example, a respondent refused to answer specific questions. Most of the derived variables for the dwelling sample, on the other hand, have no missing data, as missing values have been imputed based on cases with similar characteristics. Where imputation has taken place there are accompanying flag variables to indicate the level of imputation. Imputation is not always achievable so for some of the key derived variables there will be missing data which you may wish to distribute proportionately using the tool we have provided within the stock dataset. The method of data imputation is described more fully in the [EHS Technical Report](#) on the EHS pages on Gov.uk.
- 42 Alongside the annual data files, the UK Data Service holds supporting documentation such as questionnaires, specifications for derived variables and show cards.
- 43 Anonymisation techniques are applied to the End User Licence and Special Licence datasets released through the UK Data Service to maintain the confidentiality of respondents. This includes bottom/top coding of continuous variables to avoid outliers in the data, a process whereby values above or below an upper/lower band are capped and aggregating similar categories/responses to reduce small sample sizes. Data have also been cleaned of all identifiable variables to maintain the confidentiality of respondents, and, in a few limited cases, data swapping has taken place for disclosure control reasons.

Key concepts and definitions¹

- 44 The Housing stock dataset relates to both dwellings and households. It is therefore important to understand how these terms are defined and applied in the survey.
- 45 **Dwelling** – a unit of accommodation which may comprise one or more household spaces (a household space is the accommodation used or available for use by an individual household). A dwelling may be classified as shared or unshared. A dwelling is shared if:
- the household spaces it contains are ‘part of a converted or shared house’, or
 - not all of the rooms (including kitchen, bathroom and toilet, if any) are behind a door that only that household can use, and
 - there is at least one other such household space at the same address with which it can be combined to form the shared dwelling.

Dwellings that do not meet these conditions are unshared dwellings.

¹ A full glossary of terms used can be found in the EHS Headline and Annual Reports

The EHS definition of dwelling is consistent with the Census 2011.

- 46 **Household** – One person or a group of people (not necessarily related) who have the accommodation as their only or main residence, and (for a group) share cooking facilities and share a living room or sitting room or dining area. The EHS definition of household is slightly different from the definition used in the 2011 Census. Unlike the EHS, the 2011 Census did not limit household membership to people who had the accommodation as their only or main residence. The EHS included that restriction because it asks respondents about their second homes, the unit of data collection on the EHS, therefore, needs to include only those people who have the accommodation as their only or main residence.

Locating information in the database

Interview (household) survey variables

- 47 The topics covered in the interview survey questionnaire are illustrated in Figures 1 and 2 below, and includes a series of demographic and housing questions.
- 48 The demographic questions used to represent the Integrated Household Survey (IHS) core questions which were asked on the EHS until 2010-11. Although the EHS no longer forms part of the IHS, the questions are still asked on the 2019-20 EHS, due to their relevance to the EHS. These are listed in Figure 1.
- 49 The second part of the questionnaire is devoted to housing topics. For example there is a module of questions on mortgages (routed to households in the owner-occupied tenure) and other modules on rents, buying aspirations, and energy efficiency. These housing topics are summarised in Figure 2.

Please note that wherever variable names have remained unchanged from the former SEH or EHCS – i.e. when a question has been taken directly from one of the former surveys with the same routing, question wording and response categories – the variable name is unchanged. Where there has been a small change – e.g. a change to a response category or a slight change to the routing – then a number is added to the variable name to indicate this is a variant of the original variable e.g. Vnllrd2 – (From whom did the household rent the property?), previously this was Vnllrd.

Rotating modules

- 50 Due to pressure on the length of the interview survey, the EHS also uses the concept of 'rotating modules' to bring questions in and out of the survey to meet users' demands and policy needs. The pattern of

rotation will vary with some topics being included every other year, while others will be retained for two years in/two years out basis depending on policy interest. Full details are given on the [interview survey questionnaire](#).

- 51 Due to the nature of the rotating modules, not all topics will be covered on the housing stock dataset. This is because the housing stock requires 2 full years of data.

E.g. the second homes module was asked in 2018, but not in 2019. Therefore the housing stock dataset for 2019 (combined years 2018 and 2019) will not contain this module as there is only 1 year's worth of data.

- 52 The precise set of files and data provided each year will therefore vary reflecting the changing content of the interview survey. Details of the annual changes are available in the [EHS Technical Report](#).

- 53 The content of the physical survey remains largely unaltered year on year. Details of the annual changes are available in the [EHS Technical Report](#). The survey form for each year can be individually [downloaded](#) from the relevant EHS pages on Gov.uk.

Multi-coded variables

- 54 The EHS contains a number of multi-coded questions. These are questions where households could give more than one answer; for example 'adaptations for disability' or 'source of funding for buying a property'.

In the EHS household dataset, all multi-coded questions have been stored as a series of binary variables, with a separate variable for each answer option. It is expected that users may find this approach facilitates easier analysis. The following example (Example A) has been taken from the EHS questionnaire documentation:

Example A

Variable name

BnRec (Multicoded variable delivered as indicated below)

File name

Income.sav

Question wording

Are [you/you and your partner/spouse] receiving any of the state benefits or allowances shown on this card?

CODE ALL THAT APPLY

IF ASKED: Employment and Support Allowance is being phased in (from 1st October 2010) to replace Incapacity Benefit.

Households can currently be in receipt of Incapacity Benefit and ESA.

Binary response options

(0) No

(1) Yes

(-8) No answer

(-9) Does not apply

Multi-code answer options

(1)	Universal Credit	BnUniCrd*
(2)	Income Support/Minimum Income Guarantee	BnIncSup
(3)	Jobseekers Allowance (formerly Unemployment Benefit or Income Support for unemployed people)	BnJSA
(4)	Pension Credit	BnPenCrd
(5)	N.I. retirement pension or Old Persons Pension	BnStateP
(6)	Incapacity Benefit (previously sickness and/or invalidity benefits)	BnInCap
(7)	Employment and Support Allowance (from Q3 2008 onward)	BnESA
(8)	Child Benefit	BnChildB
(9)	Working Tax Credit (replaces Working Families and Disabled Tax Credit)	BnWTC
(10)	Child Tax Credit (replaces Children's and Childcare Tax Credit)	BnCTC
(11)	In Work Credit (from Year 2 onwards)	BnIWC
(12)	Return To Work Credit (from Year 2 onwards)	BnRTWC
(13)	None of these	BnNone
(14)	Refused	BnnRF
(15)	Don't Know	BnDKnw

Figure 1: Demographic profile questions, EHS 2019-20

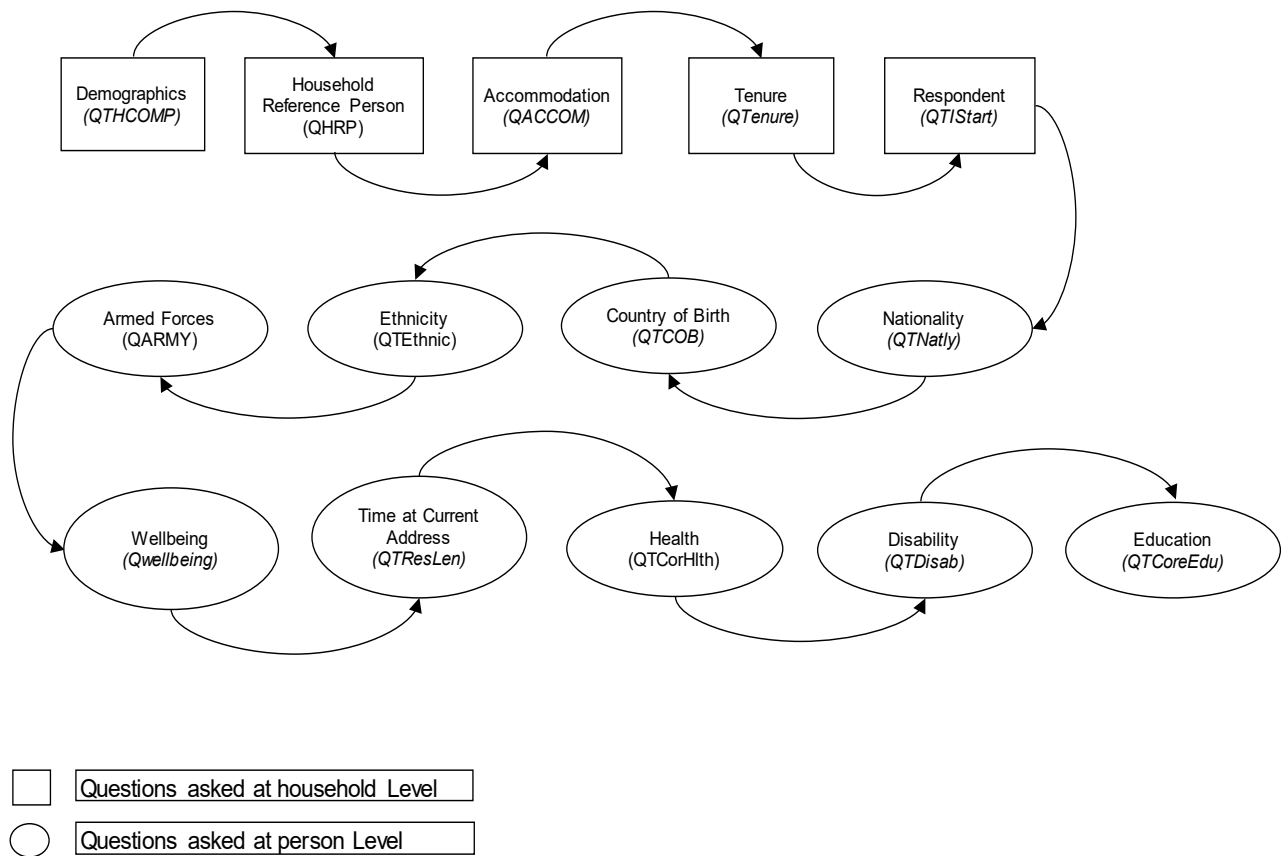
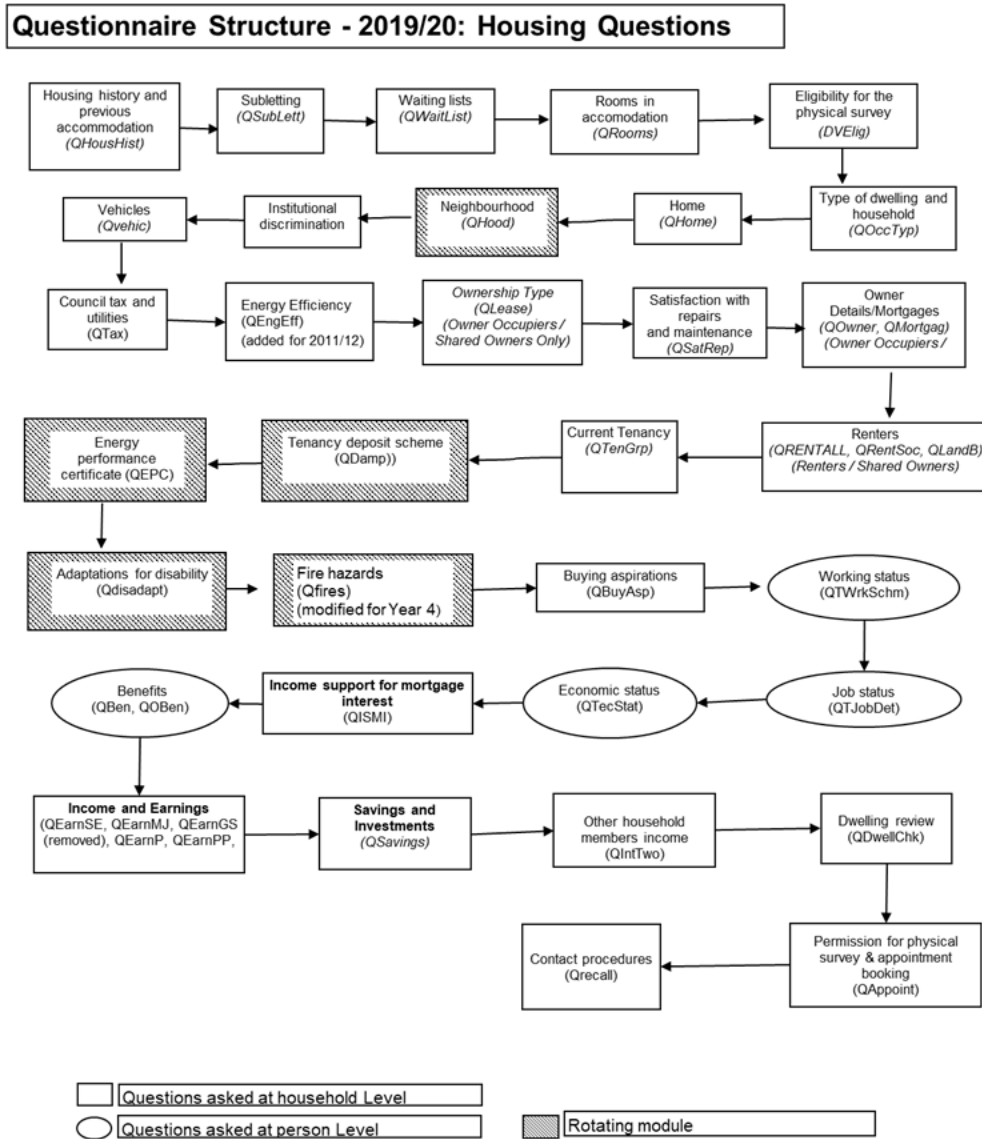


Figure 2: Housing questions, EHS 2019-20



How to match files

Matching files that exist at the household level

- 55 The key variable for matching at the dwelling/household level within or across any of the 2019 survey data is **serialanon** (an eleven digit numeric variable). You must specify this as the key field when matching. All files on the database have been sorted by this variable to support the matching of data files. 'Sort file' or 'split file' commands, however, re-sort the file. Matching with standard database files will fail if the data is not first re-sorted by serialanon.

Matching lower level files (i.e. below dwelling or household level) up to the grossing file

- 56 Files at a lower level of organisation will have more than one case per household or dwelling. Some of them have exactly the same number of cases per address e.g. the windows_SL.sav file has 7 cases for each address each representing a type of window, the chimneys_SL.sav has 2 cases per address. With other files, the number of cases varies e.g. for people_SL.sav the number of cases is the number of people in the household so one person households will just have one case, 6 person households will have 6 cases etc. As there is always more than one case per address, these cannot simply be matched into the dwelling or household level files to give results at the dwelling or household level. There are 2 ways to approach these files:
- Match the household or dwelling level files in as 'tables'. Here the information at household or dwelling level will be copied for each person etc. within that serialanon. The results will then represent the number of people, number of types of window etc. rather than the number of dwellings or number of households. This can be useful for very specific analyses, e.g. on numbers of people in non-decent dwellings.
 - Aggregate the lower level file up to address code level and then match it in with the household or dwelling level file. (Most key variables for analysis have been derived at the household level so this will not normally be necessary).

How to apply grossing factors to obtain national estimates

- 57 The EHS household dataset is based on a stratified sample with over sampling of the rented tenures to achieve large enough sample sizes to produce reliable results for the social and private rented sectors.

Grossing factors have been calculated to:

- Compensate for the design of the sample i.e. the over sampling of some dwellings and under sampling of others; and
- Take account of non-response bias: the survey response rates achieved for different groups of households and dwellings

N.B results must be based on data weighted by the relevant grossing factor to produce national estimates.

- 58 The EHS housing stock dataset comes with its own grossing factors covering the dwelling sample of cases (see above). The grossing factors can be found in the file **general xx+yy_EUL.sav** (and **general xx+yy_SL.sav** from the Special Licence Dataset). The relevant variables are:

- Dwelling grossing – **aagpdxxyy**
- Household grossing – **aagphxxyy**

Where xxyy indicates the two years across which fieldwork was conducted. For example, **aagpd1819** is the dwelling grossing factor for the 2-year dataset covering EHS 2018-19 and EHS 2019-20 i.e. for the EHS 2019 dataset.

Aagpdxxyy should be used for any analysis in which the aim is to provide estimates of dwellings and that includes physical survey data (e.g. percentage of non- decent dwellings) while **aagphxxyy** should be used for any analysis in which the aim is to provide estimates of households and that includes interview survey data (e.g. percentage of households in non-decent dwellings).

The grossing factors can only be used on 2-year dwelling sample datasets. They cannot be used on this data split into separate years.

- 59 The following number of cases and the grossed national totals are as follows:

Basic control totals for grossing dwellings and households in the housing stock datasets

Year	Dwellings - number of real cases	Dwellings - grossed counts (thousands)	Households - number of real cases	Households - grossed count (thousands)
2008	16,150	22,239	15,523	21,530
2009	16,150	22,335	15,512	21,554
2010	16,670	22,386	16,047	21,600
2011	14,951	22,754	14,386	21,918
2012	12,763	22,718	12,269	21,935
2013	12,498	23,254	12,008	22,583
2014	12,297	23,371	11,851	22,542
2015	12,351	23,543	11,955	22,657
2016	12,292	23,733	11,924	22,996
2017	12,320	23,950	11,963	23,197
2018	12,562	24,172	12,203	23,411
2019	12,300	24,414	11,974	23,662

It is vital that these figures are used as a basic check on analysis i.e. that these grossed counts are being achieved before embarking on any more detailed analysis on sub-sections of households or dwellings. In particular, if you are using more than one dataset, ensure you are not double-counting any year, by using datasets from either odd or even years (i.e. 2016 and 2018, or 2017 and 2019, not 2018 and 2019).

How to deal with missing data

Missing data conventions

- 60 The derived variable files contain no system missing values. Where values are unknown or the question was not asked, specific codes are used and the values are set to user defined missing. The conventions adopted are:
- a) Does not apply: -9 for single digit fields, -99 for 2-digit fields etc
 - b) Unknown: -8 for single digit fields, -88 for 2-digit fields etc
 - c) Not asked for a particular quarter: -2 for single digit fields, -22 for 2 digit fields etc
 - d) No partner: -7 for single digit fields, -77 for 2-digit fields etc
- 61 Users need to be aware of the implications of having missing values switched on or off within SPSS. It is useful for example to leave missing values switched on (the SPSS standard) if users wish to

establish the level of missing data in a frequency or cross tab. The SPSS output Case Processing Summary will show the level of missing data. This can be used to decide on whether the level of missing data is significant and whether it is appropriate to re-distribute missing cases.

- 62 It is particularly important to leave missing values switched on when dealing with a continuous variable e.g. costs to make a home decent. Including -99, -88 etc as a normal value (i.e. not as a missing value) would distort any outputs.

When you create any new variables, you should declare missing values as below:

Missing values var1 (-9).

This code sets up '-9' as a user defined missing so will be excluded from any cross tabs or statistics of var1.

Users should note that the use of negative values for missing data is a change from the former EHCS.

Dealing with missing data in analysis

- 63 Analysis that uses derived variables with no missing cases will have no problem with missing data. However, when undertaking analysis using other variables that have missing or unknown codes, these cases can be redistributed to produce national estimates.
- 64 Some procedures like 'cross tabulation' exclude missing cases from the table. To include these as the basis for any redistribution it is necessary to switch off the user defined missing values for each variable used. This is done by writing '**missing values Var1 (.)**.' into the SPSS syntax so that missing data will then be included in outputs.
- 65 Missing values can then be switched back on, by writing: **missing values var1 (-9)** into the SPSS syntax.

Checking if results are significant

- 66 An additional Excel spreadsheet, **EHS sig testing template**, is included to provide a quick method of checking whether changes in a particular indicator are statistically significant.

Appendix A: Content and level of interview survey files

The primary 'raw' interview survey files listed below are released under the Special Licence only.

File name	Contents	Level
AdaptationsHH_SL.sav	Information about what adaptations a household currently has inside or outside their home and if there are adaptations which are needed but not present why these modifications haven't been made.	household
AdaptationsPers_SL.sav	Information about whether a person in the household needs modifications to their home, whether the home is suitable for them, which adaptations are needed and whether they are moving to find somewhere more suitable for them,	person
Attitudes_SL.sav	Information on attitudes about current accommodation, area, and other opinion based questions	household
Contact_SL.sav	Key contact information from the survey	household
Damp_SL.sav	Problems with condensation, damp or mould in home	household
Disability_SL.sav	Information on types of disability	person
Dwelling_SL.sav	Type of accommodation Age of accommodation, housing history, Ownership type, buying aspirations	household
Employment_SL.sav	Employment information	person
Energy_SL.sav	Energy efficiency, Energy Performance Certificate.	household
Fire_SL.sav	Fires in the accommodation	household
FirstImp_IS_SL.sav	Key information on the first impressions of the dwelling and neighbourhood as recorded by interviewer	household
Hhldtype_SL.sav	Tenure, type of dwelling and household, council tax and utilities.	household
Identity_SL.sav	Country of birth, ethnicity and length of residence at current address	person
Income_SL.sav	Information on amounts and sources of income including benefits	household
Owner_SL.sav	Ownership details, mortgage	household
People_SL.sav	Key info for each person (age, sex etc.) and relationships with other people in the household	person
Renter_SL.sav	Renting, social renters, housing benefit, tenancy agreements and deposits,	household
Rooms_SL.sav	Key information about the number of rooms and whether shared	household
Vacant_SL.sav	Information relating to vacant properties from the doorstep form	household
Waitlist_SL.sav	Waiting lists for social housing module	household

Appendix B: Content and level of physical survey files

The primary 'raw' physical survey files listed below are released under the Special Licence only.

File Name	Section on survey form	Level
<i>Amenity_SL.sav</i>	5. Interior – Amenities	Dwelling
<i>Around_SL.sav</i>	18. Exterior - plot of survey dwelling (not shared plots) 19. Around the house/module 20. Block 24. Local area	Dwelling
<i>Chimney_SL.sav</i>	18. Exterior of house/module (front and rear views)	Type of chimney
<i>Commac_SL.sav</i>	9. Common parts of flat surveyed: accessways (common parts by accessway, 3 cases per address)	Type of accessway
<i>Common_SL.sav</i>	9. Common parts of flat surveyed	Dwelling
<i>Damppc_SL.sav</i>	18. Exterior of house/module (front and rear views)	Type of damp proof course
<i>Doors_SL.sav</i>	18. Exterior of survey dwelling (front and rear views)	Type of door
<i>Dormers_SL.sav</i>	18. Exterior of house/module (front and rear views)	Type of dormer/bay
<i>Elevate_SL.sav</i>	16. Elevation features 17. Specification of views	Dwelling
<i>Firstimp_PS_SL.sav</i>	2. Dwelling identification 3. Dwelling description and occupancy 4. Module associated with address surveyed	Dwelling
<i>Flatdets_SL.sav</i>	8. Details of flat	Dwelling
<i>Hhsrs_SL.sav</i>	22. Housing Health and Safety Rating System	Dwelling
<i>HQ_SL.sav</i>	7. Household questionnaire conducted by surveyor	Dwelling
<i>Interior_SL.sav</i>	5. Interior 18. Exterior of house/module (conservatories)	Dwelling
<i>Introoms_SL.sav</i>	5. Interior	Rooms
<i>Numflats_SL.sav</i>	10. Number of flats in the module	Dwelling
<i>Plotlvi_SL.sav</i>	18. Exterior - plot of survey dwelling (not shared plots)	Plot
<i>Roofcov_SL.sav</i>	18. Exterior of house/module (front and rear views)	Type of roof covering
<i>Rooffeat_SL.sav</i>	18. Exterior of house/module (front and rear views)	Type of roof feature
<i>Roofstru_SL.sav</i>	18. Exterior of house/module (front and rear views)	Type of roof structure
<i>Services_SL.sav</i>	5. Interior - Primary services 5. Interior - Space heating 6. Loft inspection	Dwelling
<i>Shape_SL.sav</i>	12. House/module shape 13. External dimension of house/module 14. Material and construction of house/module 15. Improvements/alterations and when done	Dwelling
<i>Shared_SL.sav</i>	11. Shared facilities and services	Dwelling
<i>Structure_SL.sav</i>	21. Structural defects	Dwelling
<i>Wallfin_SL.sav</i>	18. Exterior of house/module (front and rear views)	Type of wall finish
<i>Wallstru_SL.sav</i>	18. Exterior of house/module (front and rear views)	Type of wall structure
<i>Windows_SL.sav</i>	18. Exterior of survey dwelling (front and rear views)	Type of window

Appendix C: Content of derived files

The general, interview and physical derived files are released under both the End User Licence and Special Licence.

The End User Licence version of these datasets contain fewer variables than the Special Licence. For example, both the End User Licence and the Special Licence derived interview data file include variables on the age of the Household Reference Person (HRP) grouped into bands, then in addition, the Special Licence includes a continuous variable on the age of the HRP. For further detail, a list of the general, interview and physical derived variables released under the End User Licence and Special Licence is available in the EHS Technical Report (Chapter 5).

<i>Name of derived file</i>	<i>Variables</i>
<i>General 18+19_SL.sav (Special Licence)</i> <i>General 18+19_EUL.sav (End User Licence)</i>	General variables such as grossing factors, tenure and region
<i>Interview 18+19_SL.sav (Special Licence)</i> <i>Interview 18+19_EUL.sav (End User Licence)</i>	Derived variables from the interview survey, eg household composition, household income and length of residence
<i>Physical 18+19_SL.sav (Special Licence)</i> <i>Physical 18+19_EUL.sav (End User Licence)</i>	Derived variables from the physical survey, eg dwelling age, dwelling type, SAP rating and decent homes

The detailed derived files listed below are released under the Special Licence only.

<i>Name of detailed derived file</i>	<i>Variables</i>
<i>18+19 paired sample equivalised income_SL.sav</i>	Variables based on equivalised income
<i>Actual costs 18+19_SL.sav</i>	Variables for external and internal works
<i>Dimensions 18+19_SL.sav</i>	Number of rooms, wall and floor areas
<i>Energy performance 18+19_SL.sav</i>	SAP ratings, energy efficiency ratings, light, water and space heating costs, energy upgrade costs
<i>Standardised costs 18+19_SL.sav</i>	All costs per square metre of floor area
<i>HHSRS 18+19_SL.sav</i>	Variables based on 26 hazards